

10-9

Theorem 12.1 (QR-factorization) Let $A \in \mathbb{R}^{n \times m}$ have linearly independent columns $\{\vec{v}_1, \dots, \vec{v}_m\}$ and let $\{\vec{q}_1, \dots, \vec{q}_m\}$ be an orthonormal set obtained by G-S ($\vec{v}_k \in \text{span}(\vec{q}_1, \dots, \vec{q}_k)$)

Then $A = QR$ where

$$Q = \begin{bmatrix} \vec{q}_1 & \dots & \vec{q}_m \end{bmatrix}_{n \times m}$$

and $R = \begin{bmatrix} \langle \vec{v}_1, \vec{q}_1 \rangle & \langle \vec{v}_2, \vec{q}_1 \rangle & \dots & \langle \vec{v}_m, \vec{q}_1 \rangle \\ & \langle \vec{v}_2, \vec{q}_2 \rangle & \dots & \langle \vec{v}_m, \vec{q}_2 \rangle \\ & & \ddots & \vdots \\ & & & \langle \vec{v}_m, \vec{q}_m \rangle \end{bmatrix}_{m \times m}$

$\swarrow$
invertible

This is the reduced QR factorization of A and is unique.

Orthogonal Projections Revisited

Given $A = QR$ as above (i.e. the reduced QR factorization of A) we have that the orthogonal projection onto range A is given by

$$\begin{aligned} P_A &= A(A^T A)^{-1} A^T = QR(\underbrace{R^T Q^T Q R}_{I})^{-1} R^T Q^T \\ &= QR(R^T R)^{-1} R^T Q^T \\ &= QRR^{-1}(R^T)^{-1} R^T Q^T \\ &= \underline{\underline{QQ^T}} \end{aligned}$$

Also the least square solution to $A\vec{x} = \vec{b}$ is given by $\hat{\vec{x}} = (A^T A)^{-1} A^T \vec{b} = R^{-1} Q^T \vec{b}$.

Exercise:

1) Find the reduced QR factorization of the matrix $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$ and use it to

find the vector in the range of A that is closest to $\begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}$

Letting $\vec{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ and $\vec{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$ and applying G-S we get

$$\vec{e}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$$\begin{aligned} \vec{x}_2 &= \vec{v}_2 - \underbrace{\langle \vec{v}_2, \vec{e}_1 \rangle}_{\frac{1}{\sqrt{2}}} \vec{e}_1 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \\ &= \frac{1}{2} \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} \end{aligned}$$

$$\vec{e}_2 = \frac{1}{\sqrt{6}} \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}$$

Now $A = QR$ where $Q = \frac{1}{\sqrt{6}} \begin{bmatrix} \sqrt{3} & -1 \\ 0 & 2 \\ \sqrt{3} & 1 \end{bmatrix}$

and $R = \begin{bmatrix} \langle \vec{v}_1, \vec{e}_1 \rangle & \langle \vec{v}_2, \vec{e}_1 \rangle \\ 0 & \langle \vec{v}_2, \vec{e}_2 \rangle \end{bmatrix}$

$$= \begin{bmatrix} \sqrt{2} & 1/\sqrt{2} \\ 0 & 3/\sqrt{6} \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 2 & 1 \\ 0 & \sqrt{3} \end{bmatrix}$$

Now the vector in range A closest to $\begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}$ is given by

$$\begin{aligned}
 P_A \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix} &= Q Q^T \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix} \\
 &= \frac{1}{6} \begin{bmatrix} \sqrt{3} & -1 \\ 0 & 2 \\ \sqrt{3} & 1 \end{bmatrix} \begin{bmatrix} \sqrt{3} & 0 & \sqrt{3} \\ -1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix} \\
 &= \frac{1}{6} \underbrace{\begin{bmatrix} 4 & 2 & 2 \\ 2 & 4 & -2 \\ 2 & -2 & 4 \end{bmatrix}}_{P_A} \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix} \\
 &= \frac{1}{6} \begin{bmatrix} 14 \\ -8 \\ 22 \end{bmatrix}
 \end{aligned}$$

Non uniqueness of LU and QR

If A is singular (not invertible) then LU need not be unique

Example: Given singular matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & \lambda & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

for any λ .

The (non reduced) QR factorization need not be unique

Example: Given $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$ we have that

$$A = QR \text{ where } Q = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{6} \\ 0 & 2/\sqrt{6} \\ 1/\sqrt{2} & 1/\sqrt{6} \end{bmatrix} \pm \vec{e}_1 \times \vec{e}_2$$

$\underbrace{\hspace{10em}}_{Q_1}$

$$R = \begin{bmatrix} \sqrt{2} & 1/\sqrt{2} \\ 0 & 3/\sqrt{6} \\ 0 & 0 \end{bmatrix} \} R_1. \text{ Here } Q_1 R_1 \text{ is the reduced}$$

QR factorization.

Determinants

Let $T \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^n)$ and let $A = [T]_{\{\vec{e}_i\}}$. Given some subset $S \subseteq \mathbb{R}^n$ we define the image of S under T as $T(S) = \{ \vec{y} : \vec{y} = T\vec{x} \text{ for some } \vec{x} \in S \}$.

$$\begin{aligned} \text{Denote } I^n &= \{ \vec{x} \in \mathbb{R}^n : 0 \leq x_i \leq 1 \text{ for } 1 \leq i \leq n \} \\ &= [0, 1] \times \cdots \times [0, 1] \end{aligned}$$

to be the unit n -cube in $\mathbb{R}^n$.

Theorem 12.2 If $T \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^n)$, then there exists $D \in \mathbb{R}_+$ such that for any $S \subseteq \mathbb{R}^n$

$$\text{Vol}(T(S)) = D \cdot \text{Vol}(S)$$

Compare with change of variables formula for multiple integrals

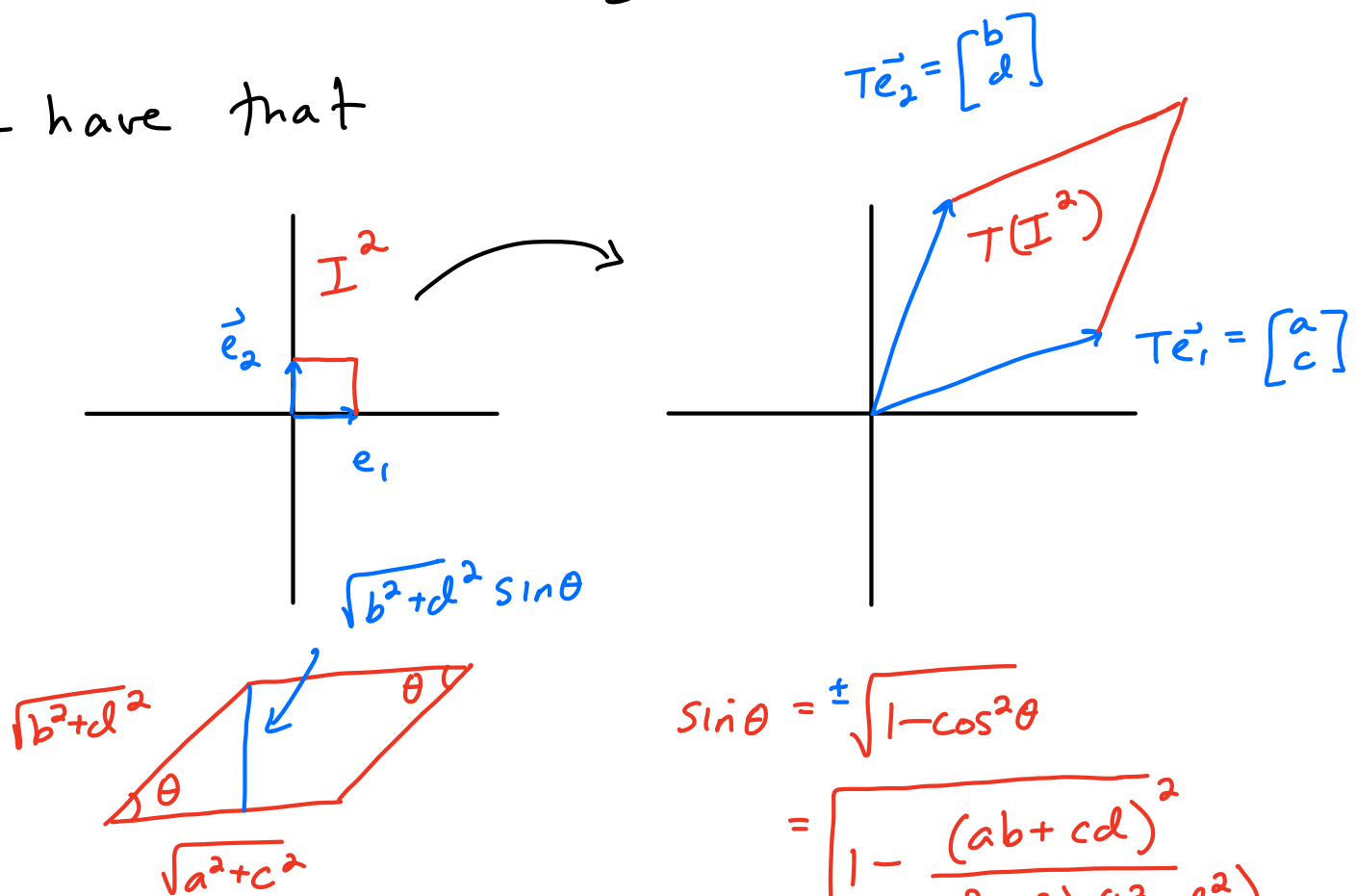
"Proof": let I^n be the unit n -cube in $\mathbb{R}^n$ and let $A = [T]_{\{\vec{e}_i\}}$. Now taking the SVD (singular value decomposition) of A we have that $A = UDV$ where U and V are orthogonal and D is diagonal. Now $\text{vol}(T(I^n)) = \prod_{i=1}^n D_{i,i} \text{Vol}(I^n)$ and the theorem from the linearity of T and translation invariance of volume.

Def 12.3 Given T and A as above we define $\det A = \text{sgn}(A) \cdot \text{vol}(T(I^n))$ where $\text{sgn}(A) = (-1)^n$ where n is the # of reflections in the QR factorization. ($\text{sgn}(A) = \det Q$)

2×2 Determinant

letting $A = [T]_{\{\vec{e}_i\}} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $I^2 = [0,1] \times [0,1]$

we have that



$$\begin{aligned} \sin \theta &= \pm \sqrt{1 - \cos^2 \theta} \\ &= \sqrt{1 - \frac{(ab+cd)^2}{(a^2+c^2)(b^2+d^2)}} \\ &= \frac{ad-bc}{\sqrt{a^2+c^2} \sqrt{b^2+d^2}} \end{aligned}$$

$$\begin{aligned} \text{and } \text{Vol}(T(I^2)) &= \sqrt{a^2+c^2} \sqrt{b^2+d^2} \sin \theta \\ &= \pm (ad-bc) \end{aligned}$$

Laplacian expansion

$$\begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}$$

$$\det A = |A| = \sum_{j=1}^n (-1)^{i+j} A_{ij} |M_{ij}|$$

(expanding across the i th row)

where M_{ij} is the minor matrix obtained by removing the i^{th} row and j^{th} column of A .

Theorem 12.4 Let $A, B \in \mathbb{R}^{n \times n}$. Then

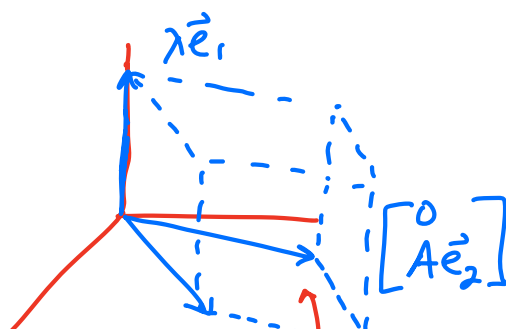
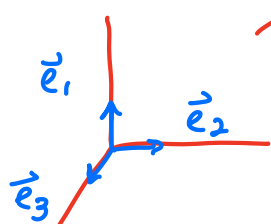
$$\det(AB) = (\det A)(\det B)$$

"Proof": Let I^n be the unit n -cube and assume that $\det B \neq 0$. Now WLOG assuming that $\text{sgn}(A) = \text{sgn}(B) = 1$ we have that from theorem 12.2

$$\begin{aligned} \det(AB) &= \text{Vol}((A \circ B)(I^n)) \frac{\text{Vol}(B(I^n))}{\text{Vol}(B(I^n))} \\ &= \frac{\text{Vol}(A(B(I^n)))}{\text{Vol}(B(I^n))} \cdot \text{Vol}(B(I^n)) \\ &= \det A \cdot \det B \end{aligned}$$

Theorem 12.5 The block diagonal matrix

$$B = \begin{bmatrix} \lambda & 0 & \dots & 0 \\ 0 & & & \\ \vdots & & A & \\ 0 & & & \end{bmatrix} \text{ has determinant } \det B = \lambda \cdot \det A$$



$$\begin{bmatrix} 0 \\ A\vec{e}_3 \end{bmatrix} \quad \det A$$

Corollary 12.6 Given the block diagonal matrix

$$A = \begin{bmatrix} A_1 & & \\ & \ddots & \\ & & A_n \end{bmatrix} = A_1 \oplus \dots \oplus A_n, \quad \det A = \prod_{i=1}^n \det A_i$$

Exercise:

2) Compute the following determinants

a. $\rightarrow \begin{vmatrix} 3 & 7 & -2 \\ -4 & 1 & 6 \\ 2 & 0 & -3 \end{vmatrix}$

$$\begin{aligned} &= 2 \begin{vmatrix} 7 & -2 \\ 1 & 6 \end{vmatrix} - 3 \begin{vmatrix} 3 & 7 \\ -4 & 1 \end{vmatrix} \\ &= 2(44) - 3(24) \\ &= 16 \end{aligned}$$

b. $\rightarrow \begin{vmatrix} 1 & 4 & -8 & 9 \\ 0 & 0 & 3 & 0 \\ 5 & 1 & 2 & 1 \\ 2 & 0 & 5 & 0 \end{vmatrix}$

$$\begin{aligned} &= -3 \begin{vmatrix} 1 & 4 & 9 \\ 5 & 1 & 1 \\ 2 & 0 & 0 \end{vmatrix} \\ &= -3 \cdot 2 \begin{vmatrix} 4 & 9 \\ 1 & 1 \end{vmatrix} \\ &= 30 \end{aligned}$$

12.7 Determinants of Special matrices

i. Identity: $\det I = 1$

ii. Diagonal matrix: If D is diagonal, then
 $\det D = \prod D_{i,i}$

iii. Triangular matrices: If U and L are upper and lower triangular matrices, then

$$\det U = \prod U_{i,i}$$

$$\det L = \prod L_{i,i}$$

iv. Elementary matrices: If $E_{ij}(\lambda_{ij}) = I + \begin{bmatrix} & \lambda_{ij} \\ & \end{bmatrix}$ where $i \neq j$ then $\det E_{ij} = 1$.

v. Permutation matrices: Given permutation σ , $\det P_\sigma = (-1)^{\text{sgn}(\sigma)} = (-1)^{\# \text{ row swaps}}$

Example: For $\sigma = (12)$ that takes $\sigma(1)=2$, $\sigma(2)=1$ and $\sigma(3)=3$ we have that $P_\sigma = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ where

$$P_\sigma \begin{bmatrix} A_{1,\cdot} \\ A_{2,\cdot} \\ A_{3,\cdot} \end{bmatrix} = \begin{bmatrix} A_{2,\cdot} \\ A_{1,\cdot} \\ A_{3,\cdot} \end{bmatrix} \text{ and}$$

$$\begin{bmatrix} A_{\cdot,1} & A_{\cdot,2} & A_{\cdot,3} \end{bmatrix} P_\sigma = \begin{bmatrix} A_{\cdot,2} & A_{\cdot,1} & A_{\cdot,3} \end{bmatrix}$$

$$\text{Here } \det P_\sigma = -1$$

12.8 Properties of Determinants

i. A is invertible iff $\det A \neq 0$

ii. If A is invertible, then $\det A^{-1} = \frac{1}{\det A}$

iii. $\det A^T = \det A$

iv. Linearity in the columns (rows)

$$\begin{aligned} & \left| \vec{v}_1 \cdots \vec{v}_{k-1} \underbrace{\sum c_i \vec{w}_i}_{\vec{v}_k} \vec{v}_{k+1} \cdots \vec{v}_n \right| \\ &= \sum_i c_i \left| \vec{v}_1 \cdots \vec{v}_{k-1} \vec{w}_i \vec{v}_{k+1} \cdots \vec{v}_n \right| \end{aligned}$$

v. $\det(\lambda A) = \det(\lambda I) \det A = \lambda^n \det A$

vi. Product of pivots: Let $PA = LDU$,

then $\det A = (-1)^r \prod D_{i,i}$ where r is

r is the # of row exchanges ($\det P$)